SOH CAH TOA and missing sides

Day 21 • 2026-12-03 • 2:15 PM–3:30 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 CHOOSE THE RATIO

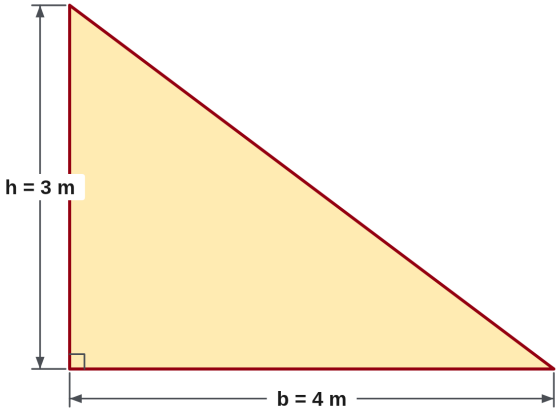
For the angle opposite the 3 m leg in this Shape Canvas triangle, which leg is adjacent?

A. The 4 m leg

B. The 3 m leg

C. The hypotenuse

D. Neither leg



Shape Canvas

Choice (A–D) _____

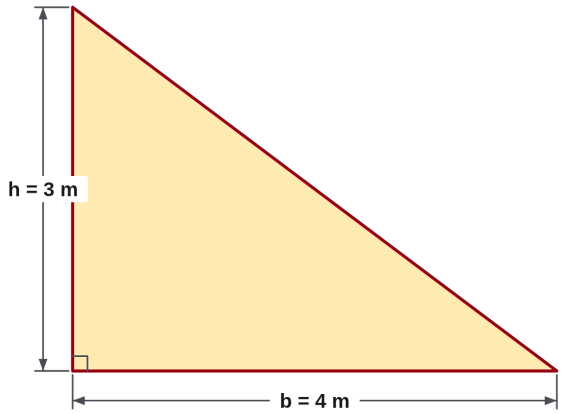
Working / reason _____

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

SOH CAH TOA and missing sides

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02 For the angle opposite the 3 m leg, find opposite divided by adjacent.



Shape Canvas

Answer

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

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03 A Numeracy Hub calculator will evaluate sin(30 degrees). Which angle mode should be selected?

A. GRAD without conversion

B. Percent mode

C. DEG

D. RAD



Numeracy Hub

Choice (A–D)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

04 Which ratio uses opposite and hypotenuse?

A. Perimeter

B. Sine

C. Cosine

D. Tangent

Choice (A–D)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

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05 Which ratio uses adjacent and hypotenuse?

A. Cosine
B. Sine
C. Tangent
D. Area

Choice (A–D)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

06 FIND THE SIDE

A right triangle has a 10 m hypotenuse and a 30-degree angle. Find the opposite side. Round to 2 decimal places.

Answer (m)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

07 A 12 m hypotenuse makes a 60-degree angle with the adjacent leg. Find that adjacent leg. Round to 2 decimal places.

Answer (m)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

08 A right triangle has an adjacent run of 4 m and an angle of 35 degrees. Find the opposite rise. Round to 2 decimal places.

Answer (m)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

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09 The opposite side is 3 m and the angle is 30 degrees. Find the hypotenuse. Round to 2 decimal places.

Answer (m) | **Working / reason**

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

10 The adjacent side is 5 m and the angle is 40 degrees. Find the hypotenuse. Round to 2 decimal places.

Answer (m) | **Working / reason**

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

11 CHECK THE SETUP

A learner needs the opposite side and knows the adjacent side and angle. Which equation is suitable?

A. Opposite = adjacent × tan(angle)
B. Opposite = adjacent × cos(angle)
C. Opposite = adjacent + angle
D. Opposite = adjacent ÷ sin(angle)

Choice (A–D) | **Working / reason**

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

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12 A calculated leg exceeds the stated hypotenuse. What is the best response?

A. Accept it because a calculator gave it

B. Rename the leg as area

C. Round it until it is smaller

D. Recheck the ratio, rearrangement and angle mode

Choice (A–D)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

13 A 10 m hypotenuse at 30 degrees gave an opposite side of 5 m. What ratio checks this result?

Answer

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

14 Which is the better first step for an unfamiliar right-triangle question?

A. Measure the drawing with a ruler

B. Mark the angle and label opposite, adjacent and hypotenuse

C. Press sine before reading the drawing

D. Add every visible number

Choice (A–D)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

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15 Why should intermediate trig values be kept unrounded?

A. Early rounding can change the final length

B. It changes a non-right triangle into a right triangle

C. It removes the need for units

D. It guarantees a whole-number answer

Choice (A–D)

Working / reason

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58